

DTSA 5727

# Computational Bayesian Statistics for Data Science

|                                                     |                                        |               |
|-----------------------------------------------------|----------------------------------------|---------------|
| FORMAT                                              | CATEGORY                               | CREDITS       |
| Online (DTSA)                                       | Elective                               | 1             |
| SECTION                                             | BEST ROLE MATCH                        | AVG ALIGNMENT |
| Section 4 – Elective                                | Machine Learning Engineer (31.2 / 100) | 23.0 / 100    |
| SPECIALIZATION Bayesian Statistics for Data Science |                                        |               |

23

CURRENT AVG SCORE  
across 6 job roles

58

POTENTIAL SCORE  
with recommended additions

+45

MAX SCORE GAIN  
best single-role improvement

CURRENT COURSE DESCRIPTION

This course introduces students to the theoretical underpinnings and applications of computational algorithms in Bayesian statistics. Topics include maximum a posteriori estimation; rejection sampling; and Markov chain Monte Carlo algorithms, such as the Gibbs sampler and several variations of the Metropolis-Hastings algorithm. This course also provides an introduction to the Stan computing environment.

INDUSTRY ALIGNMENT BY JOB ROLE

| JOB ROLE                  | CURRENT SCORE          |      | PROJECTED SCORE        |      | POINT GAIN | PRIORITY |
|---------------------------|------------------------|------|------------------------|------|------------|----------|
| Machine Learning Engineer | <div><div></div></div> | 31.2 | <div><div></div></div> | 58.8 | +27.6      | MEDIUM   |
| Applied Scientist         | <div><div></div></div> | 27.4 | <div><div></div></div> | 72.1 | +44.7      | HIGH     |
| Data Scientist            | <div><div></div></div> | 26.3 | <div><div></div></div> | 66.8 | +40.6      | HIGH     |
| AI Engineer               | <div><div></div></div> | 22.1 | <div><div></div></div> | 54.1 | +32.0      | HIGH     |
| ML Engineer               | <div><div></div></div> | 20.1 | <div><div></div></div> | 42.3 | +22.2      | MEDIUM   |
| MLOps Engineer            | <div><div></div></div> | 11.0 | <div><div></div></div> | 55.4 | +44.4      | HIGH     |

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold ( **example** ) appear as gaps across 3 or more job roles.

System Design

3 of 6 roles

systems

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

HIGH

Add System Design coverage — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

GAP KEYWORDS BY ROLE

|                                                                               |                         |
|-------------------------------------------------------------------------------|-------------------------|
| Machine Learning Engineer                                                     | 31.2 → 58.8 (+27.6 pts) |
| data sets learning tools design development science models                    |                         |
| Applied Scientist                                                             | 27.4 → 72.1 (+44.7 pts) |
| research interdisciplinary problems models technical department data learning |                         |
| Data Scientist                                                                | 26.3 → 66.8 (+40.6 pts) |
| data sets presentation social problems                                        |                         |
| AI Engineer                                                                   | 22.1 → 54.1 (+32.0 pts) |
| data models learning technical systems teams problems machine                 |                         |
| ML Engineer                                                                   | 20.1 → 42.3 (+22.2 pts) |
| data learning machine systems models world problems new                       |                         |
| MLOps Engineer                                                                | 11.0 → 55.4 (+44.4 pts) |
| data scientists systems learning machine tools models development             |                         |

**Methodology:** Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.